Source-Based Interaction Modeling Captures Traffic Pollution Redistribution That Static Estimation Cannot

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Abstract

Street-segment surfaces of traffic NO2 and noise are increasingly used as spatial data products, yet most are produced by static estimation that regresses pollution onto fixed geographic predictors such as land use, road class, and population density. Such estimators have a blind spot for spatial systems: their inputs do not depend on the state of traffic on the network, so the most basic network what-if, removing an edge to close a street, leaves their predicted surface unchanged. We present an agent-based model in which individual vehicles interact on an OSMnx street network, following one another, queuing at signals, and backing up in congestion, and map the resulting simulated traffic to per-segment NO2 (HBEFA emission factors) and noise (the EU CNOSSOS method). Under an edge-removal (road-closure) experiment, the agent model redistributes on average 61 to 76 percent of a closed arterial's NO2, depending on the arterial, onto the parallel routes that detouring traffic takes up. The contribution is a demonstration of capability, not a claim of greater accuracy: source-based interaction modeling captures the spatial redistribution of traffic pollution that static estimation, by the nature of its inputs, cannot.

CCS Concepts

• **Computing methodologies** → **Agent / discrete models**; • **Information systems** → *Geographic information systems.*

Keywords

road networks, what-if and counterfactual analysis, spatial surfaces, agent-based modeling, traffic simulation, air quality, environmental noise

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1 Introduction

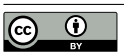
Fine-scale urban surfaces of NO2 and road-traffic noise are increasingly treated as spatial data products, per-segment maps that downstream systems query for routing, facility siting, and exposure analysis. Most are produced by static estimation: a model (land-use regression, or a land-use random forest) is fit on fixed geographic predictors, land use, road classification, population density, distance to major roads, and predicts pollution at street-segment resolution. These estimators are accurate where conditions are stable and cheap to deploy.

They share one structural limitation that matters specifically for spatial systems: their predictors describe the fixed configuration of the city, not the state of traffic on the network. Consider the most basic counterfactual a spatial system can pose: remove an edge (close a street) and ask how the pollution map responds. Because none of the estimator's land-use inputs depend on the network's traffic state, its prediction is provably invariant under the edge removal: the surface cannot change in response to the very perturbation that changed the traffic.

We show that source-based interaction modeling supplies what the static surface lacks, and we make the claim falsifiable with an edge-removal experiment on a real road network: an agent-based model (ABM) of vehicles that follow, queue, and congest on the Portland arterial network, from which we derive per-segment NO2 and noise surfaces and compare against a strong static baseline under the same closure. Static estimation is not the only alternative to a source-based surface: dynamic chains that drive emission and dispersion models from microscopic or mesoscale traffic already exist [8, 9]. It is, however, the family that produces the per-segment surfaces now published as data products, and the family with the invariance property above. Our contribution is therefore not the chain but the spatial comparison against static estimation, and the demonstration of a redistribution effect, traffic and its pollution moving onto parallel routes when an edge is cut.

2 Method

Network and vehicle dynamics. We model individual vehicles on an OSMnx street network [1] of the SE Powell Boulevard corridor in Portland (978 nodes, 2,838 directed edges). Vehicles take origin-destination routes that minimize travel time and advance segment by segment under an intelligent-driver car-following kernel [10], adjusting speed to the vehicle ahead. Thirty percent of trips are regional through-traffic, entering the study area at one network edge and leaving at another rather than starting and ending inside the corridor; we assign those trips to arterials from street geometry and road class, not by tuning against the held-out traffic counts. Vehicles queue at 21 OSM-tagged signalized intersections under a two-phase cycle, and queues spill back across intersections: a vehicle with no leader on its own segment brakes for the queue on the next one and holds at the stop line. From these interactions congestion emerges: mean speed drops and queues stack, the behavior that justifies an ABM rather than a static flow assignment.



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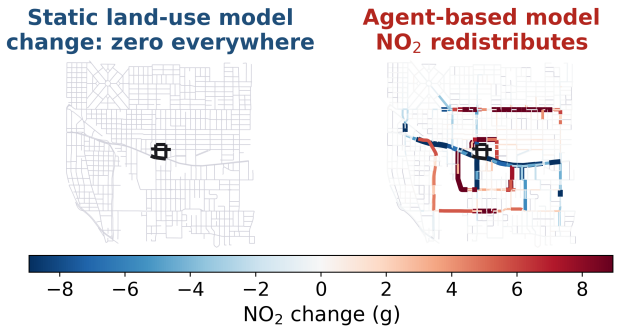


Figure 1: Close SE Powell and only the agent-based model responds. Same closure, one shared change scale: red up, blue down, grey unchanged; the black block marks the closed segments. The closure changes none of the static model's land-use inputs, so its predicted change is zero on every segment, while the agent model drops NO2 about 65% on the closed corridor and raises parallel SE Division by 41% and SE Holgate by 57%.

Emissions and noise. From each vehicle's instantaneous speed and acceleration we accumulate NOx per segment using an HBEFA-derived NOx(v, a) factor [3]. The NO2 surface is reported as a fixed NO2/NOx fraction applied in post-processing. The same simulated traffic drives a second surface: per-segment road-traffic noise in dB(A), computed with the A-weighted EU CNOSSOS method [2] and made congestion-aware by recovering realized mean speed per segment.

3 Experiment and Results

Two checks show that neither side of the comparison is a strawman. The simulated per-segment volumes rank-correlate with held-out real traffic counts at Spearman 0.59, and the static baseline, a random forest fit on land-use and network predictors, reaches an out-of-bag R squared of 0.71 on the log scale of its fit. In line with prior work on structure-based traffic prediction [4], the remaining spatial gap is attributable to the absence of real origin-destination demand, which the closure result does not depend on closing.

We then pose the edge-removal counterfactual: close a 150 m zone on the arterial (24 directed segments) and rerun the same demand model and random seed so vehicles reroute around it. The static surface does not move at all, because none of its land-use inputs change. The agent model responds the way the mechanism predicts: NO2 leaves the closed arterial and reappears on the parallel routes that detouring traffic takes up. Closing an arterial strips a mean of 61 to 76 percent of its own NO2, depending on the arterial; in the SE Powell case, the corridor-wide NO2 drops about 65 percent while parallel SE Division and SE Holgate rise by about 41 and 57 percent. Network-total NO2 moves only slightly, by about 5 percent, because the detours are longer and slower, so the effect is redistribution rather than a change in magnitude.

The result holds across twelve random seeds and three different arterials, and corresponds to a shift in modeled NO2 exposure for roughly 11,000 residents when the surface is intersected with block-group population.

4 Discussion, Limitations, and Ongoing Work

The contribution is a demonstration of capability, not a claim of greater absolute accuracy. Portland has sparse pollution ground truth, only two regulatory NO2 monitors in the metro, so we frame this honestly as a comparison between methods rather than against measurement.

Current simplifications await calibration: an assumed uniform signal cycle (Powell runs adaptive SCATS control, with no published fixed plan), a single emission class, and an NO2/NOx fraction taken from the literature range. Ongoing work compares the agent-generated NO2 surface against a published land-use random-forest baseline for the same city [6], holding the statistical method fixed and the feature footing common (per-segment predictors aggregated over 100 to 1200 m buffers) so that only the predictor source differs, under spatial block cross-validation [7]; the noise surface is checked against the U.S. FHWA Traffic Noise Model [5] as a mechanistic reference.

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